

# Week 6 Practical Worksheet – Questions

## Part 1 – UML and Object-Oriented Design

- Q1. What is UML and why is it useful when designing software systems?
- Q2. In UML, what does a class represent?
- Q3. Name two components typically shown inside a UML class box.
- Q4. What is the difference between attributes and methods in a class?
- Q5. What type of relationship exists between TravelClass and Passenger?
- Q6. What type of relationship exists between Department and Crew?
- Q7. In the UML diagram, which class contains the foreign key linking to TravelClass?
- Q8. Why is a foreign key important in relational databases?

## Part 2 – Python Classes and Database Tables

- Q9. What is the purpose of the create\_table() method in the TravelClass Python class?
- Q10. What does the following SQL statement do?

```
CREATE TABLE IF NOT EXISTS class_table (  
    class_id INTEGER PRIMARY KEY AUTOINCREMENT,  
    class_name TEXT NOT NULL UNIQUE  
);
```

- Q11. What is the purpose of the add\_class() method?
- Q12. What type of method is create\_table() in the Python class?
- Q13. Why are static methods useful for database operations?
- Q14. Which Python module is used to connect to SQLite databases?
- Q15. What does the commit() function do in SQLite?

## Part 3 – SQL Queries and Data Extraction

- Q16. Write an SQL query to retrieve all passengers from the passenger table.

Q17. Write an SQL query to retrieve the names of passengers and their travel class.

Q18. What SQL command is used to combine rows from two tables?

Q19. Write a JOIN query that displays passenger names and their class.

Q20. Why is it necessary to join the passenger and class\_table tables?

#### **Part 4 – Data Preparation for Machine Learning**

Q21. Why is SQL used to filter data before passing it to a machine learning model?

Q22. What type of machine learning algorithm is used in this lab?

Q23. What is the target variable when predicting crew department?

Q24. Name two features used to predict crew department.

Q25. Why must categorical variables such as citizenship or sex be encoded before training a machine learning model?

#### **Part 5 – Decision Trees**

Q26. What is a decision tree?

Q27. What does the root node of a decision tree represent?

Q28. What does a leaf node represent?

Q29. What is the purpose of splitting data at each node?

Q30. What Python library is used to implement decision trees in this lab?

#### **Part 6 – Interpreting Machine Learning Results**

Q31. What does the accuracy score of a machine learning model represent?

Q32. Why is it important to split data into training and testing datasets?

Q33. What is the purpose of the plot\_tree() function?

Q34. Why might a decision tree use age as the first split?

Q35. What does feature importance tell us about a model?

#### **Part 7 – Advanced Questions**

Q36. Explain the difference between CRUD functions and “wisdom functions” in this lab.

Q37. Why is it useful to store machine learning functions inside a class?

Q38. What advantage does SQL provide when selecting datasets for machine learning?

Q39. Why might a decision tree trained on poorly selected features produce weak predictions?

Q40. Suggest one improvement that could be made to the machine learning model in this lab.